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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Vehicle Service Reminder and Maintenance Tracker

Submitted by

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DECLARATION

I hereby declare that the project entitled **"Vehicle Service Reminder and Maintenance Tracker"** submitted in partial fulfillment of the requirements for the course **Java Programming** is a **record of my original work** carried out under the guidance of **Mugilan R**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Shanyuktha D

CERTIFICATE

This is to certify that the project entitled **"Vehicle Service Reminder and Maintenance Tracker"** is a **bona fide work** carried out by **Shanyuktha D B.Sc.IT-'B'**, **Information Technology, KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

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Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The Vehicle Service Reminder and Maintenance Tracker is a software system designed to help vehicle owners manage and remember their vehicle service schedules. Many people forget to service their vehicles on time, which can lead to vehicle damage, higher repair costs, and reduced performance.

This system stores important details such as vehicle number, last service date, next service due date, mileage, and maintenance history. The main purpose of this project is to remind users about upcoming vehicle services and help them maintain their vehicles in good condition.

The system allows users to add vehicle details, update service records, and track maintenance activities. It can also generate reminders when the service date is near. By using this system, users can easily monitor their vehicle condition and ensure timely servicing.

This project is useful for individual vehicle owners as well as small service centers. It improves vehicle maintenance management and reduces the chances of missing important service schedules.

1. INTRODUCTION

Vehicles are an important part of our daily life. They help people travel easily and save time. To keep a vehicle in good condition, regular maintenance and servicing are very important. If vehicle service is not done on time, it may cause mechanical problems, reduced performance, and higher repair costs.

Many vehicle owners forget their service dates or do not keep proper records of their vehicle maintenance. Because of this, vehicles may not receive timely servicing. A proper system is needed to remind users about service schedules and maintain vehicle service records.

The Vehicle Service Reminder and Maintenance Tracker is developed to solve this problem. This system helps users store vehicle details, track maintenance history, and receive reminders for upcoming services. It makes vehicle management easier and more organized.

This project aims to improve vehicle maintenance by providing a simple and efficient system for tracking service schedules and maintaining vehicle information.

1.1 Objective

The main objectives of the Vehicle Service Reminder and Maintenance Tracker are:

1. To help vehicle owners keep track of their vehicle service schedules.

2. To store important vehicle details such as vehicle number, service date, mileage, and maintenance history.
3. To provide reminders for upcoming vehicle services.
4. To reduce the chances of missing important vehicle maintenance.
5. To improve vehicle performance by ensuring timely servicing.
6. To make vehicle maintenance management simple and organized.

1.2 Overview

The Vehicle Service Reminder and Maintenance Tracker is a system developed to help vehicle owners manage their vehicle service schedules and maintenance records. Many people forget the correct time for servicing their vehicles, which may lead to poor vehicle performance and unexpected repair costs.

This system allows users to store important information such as vehicle number, service date, mileage, and maintenance details. It keeps all service records in one place so that users can easily check the history of their vehicle maintenance.

The main feature of this system is to provide reminders for upcoming vehicle services. By receiving reminders, users can service their vehicles on time and maintain them properly.

Overall, this project helps users organize vehicle information, track maintenance activities, and improve vehicle care in a simple and efficient way.

1.3 Features

The main features of the Vehicle Service Reminder and Maintenance Tracker are:

- **Add Vehicle Details** – Users can enter and store vehicle information such as vehicle number, model, and owner name.
- **Service Record Management** – The system allows users to add and update vehicle service details and maintenance history.
- **Service Reminder** – The system reminds users when the next vehicle service date is near.
- **Mileage Tracking** – Users can record the vehicle mileage to know when service is required.
- **Maintenance History** – The system stores all previous service records for easy reference.

- **Friendly Interface** – The system is simple and easy for users to operate.User

2. PROGRAM ANALYSIS

Program analysis is the process of studying the system requirements and understanding how the program should work before developing it. It helps identify the problems in the existing system and find suitable solutions.

In the current situation, many vehicle owners do not maintain proper records of their vehicle services. They may forget the next service date or lose service information. This can lead to delayed maintenance and vehicle problems.

The Vehicle Service Reminder and Maintenance Tracker analyzes these issues and provides a solution by creating a system that stores vehicle details, tracks service history, and reminds users about upcoming services.

The system is designed to make vehicle maintenance management easier, more organized, and more efficient for users.

2.1 Problem Definition

Many vehicle owners do not keep proper records of their vehicle maintenance and service schedules. Because of this, they may forget the correct time for servicing their vehicles.

Missing regular vehicle service can cause problems such as reduced vehicle performance, engine damage, and higher repair costs. In the existing system, most people depend on memory or manual records, which are not always reliable.

Therefore, there is a need for a system that can store vehicle details, track maintenance records, and remind users about upcoming services. The Vehicle Service Reminder and Maintenance Tracker is developed to solve this problem by helping users manage their vehicle service schedules in a simple and organized way.

2.2 Project Overview

The Vehicle Service Reminder and Maintenance Tracker is a software project developed to help vehicle owners manage their vehicle service schedules and maintenance records in an easy way.

This system allows users to store important vehicle information such as vehicle number, model, service date, mileage, and maintenance details. It keeps all service records in one place so that users can easily check the history of their vehicle services.

The main purpose of this project is to remind users about upcoming vehicle services and help them maintain their vehicles properly. By using this system, users can avoid missing important service dates and ensure their vehicles remain in good condition.

Overall, the project provides a simple and effective way to manage vehicle maintenance and improve vehicle care.

2.3 Scopes and Limitations

Scope

The scope of the Vehicle Service Reminder and Maintenance Tracker includes the following:

• Storing vehicle details

• Recording service dates and maintenance history

• Tracking vehicle mileage

• Providing reminders for upcoming services

The system can be used in:

• Individual vehicle owners

• Car service centers

• Bike service stations

Limitations

• The system depends on user input for vehicle details

• Online service booking is not available

• Advanced vehicle diagnostic features are not included

• The system mainly supports basic service tracking.

3. SYSTEM DESIGN

System design describes how the system is structured and how different components work together. The Vehicle Service Reminder and Maintenance Tracker is designed to store vehicle information, manage service records, and provide reminders for upcoming services.

The system includes modules such as vehicle details management, service record management, and reminder notification. Users can add vehicle information, update service details, and view maintenance history easily.

The design focuses on a simple and user-friendly interface so that users can manage their vehicle service information efficiently.

3.1 Modules

The Vehicle Service Reminder and Maintenance Tracker system is divided into the following modules:

• **Vehicle Details Module –** This module allows users to enter and store vehicle information such as vehicle number, vehicle model, owner name, and contact details. It helps maintain all vehicle information in one place.

• **Service Record Module –** This module is used to record and update vehicle service details such as last service date, next service date, and maintenance information.

• **Reminder Module –** This module provides reminders to users about upcoming vehicle service schedules so that they do not miss important maintenance dates.

• **Maintenance History Module –** This module keeps a record of all previous vehicle services. Users can view past service details whenever required.

• **User Interface Module –** This module provides a simple and easy interface for users to interact with the system and manage vehicle information efficiently.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Vehicle Service Reminder and Maintenance Tracker, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis

• System Design

• Program Development

• Testing and Validation

• Result and Documentation

4.1. Problem Analysis

Problem analysis is the process of identifying and understanding the problems in the existing system. Many vehicle owners forget their vehicle service dates and do not maintain proper service records. This may lead to delayed maintenance, reduced vehicle performance, and higher repair costs.

The existing system mostly depends on manual records or personal memory, which is not always reliable. Because of this, there is a need for a system that can store vehicle details and remind users about upcoming services.

The Vehicle Service Reminder and Maintenance Tracker is designed to solve this problem by providing a simple system to record vehicle information, track service history, and remind users about their next service schedule.

4.2. System Design

System design explains how the system is structured and how different components work together. In the Vehicle Service Reminder and Maintenance Tracker, the system is designed to store vehicle details, manage service records, and provide reminders for upcoming services.

The design includes modules for vehicle information management, service record management, and reminder notifications. The system interface is designed to be simple and user-friendly so that users can easily manage their vehicle service information.

4.3. Program Development

Program development is the stage where the actual coding of the system is done. In this phase, the program is developed based on the system design and requirements.

The necessary programming concepts are used to create functions for storing vehicle details, updating service records, and generating service reminders. The program is designed to ensure smooth operation and easy data management.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	**Expected Result**
Enter vehicle details	Vehicle information stored correctly
Add service record	Service details saved successfully
Set next service date	Reminder scheduled correctly
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

The Vehicle Service Reminder and Maintenance Tracker system was successfully developed and tested. The system allows users to store vehicle details, manage service records, and receive reminders for upcoming vehicle services.

The system helps users keep track of their vehicle maintenance in an organized way. It reduces the chances of forgetting service dates and helps maintain vehicles properly.

The project documentation explains the system design, development process, and working of the system. It provides complete details about the features and functionality of the project.

4.6. Algorithm

1. Start the process
2. Declare variables
3. ownerName, vehicleNumber, lastServiceDate, nextServiceDate, mileage
4. Prompt and read all required inputs

5. (ownerName, vehicleNumber, lastServiceDate, nextServiceDate, mileage)

6.Validate and store the entered data into variables

7.Process the data to prepare vehicle service information

8.Display vehicle details and next service reminder using nextServiceDate

9.Stop the program

.

4.7. Flowchart



5.SOURCE CODE

```
import java.util.Scanner;

class VehicleServiceTracker {

    String vehicleNumber;
    String ownerName;
    String lastServiceDate;
    String nextServiceDate;
    int mileage;

    public void getDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter Owner Name:");
        ownerName = sc.nextLine();

        System.out.println("Enter Vehicle Number:");
        vehicleNumber = sc.nextLine();

        System.out.println("Enter Last Service Date:");
        lastServiceDate = sc.nextLine();

        System.out.println("Enter Next Service Date:");
        nextServiceDate = sc.nextLine();

        System.out.println("Enter Vehicle Mileage:");
```

```
        mileage = sc.nextInt();
    }

    public void displayDetails() {
        System.out.println("\n--- Vehicle Service Details ---");
        System.out.println("Owner Name: " + ownerName);
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Last Service Date: " + lastServiceDate);
        System.out.println("Next Service Date: " + nextServiceDate);
        System.out.println("Mileage: " + mileage);
    }

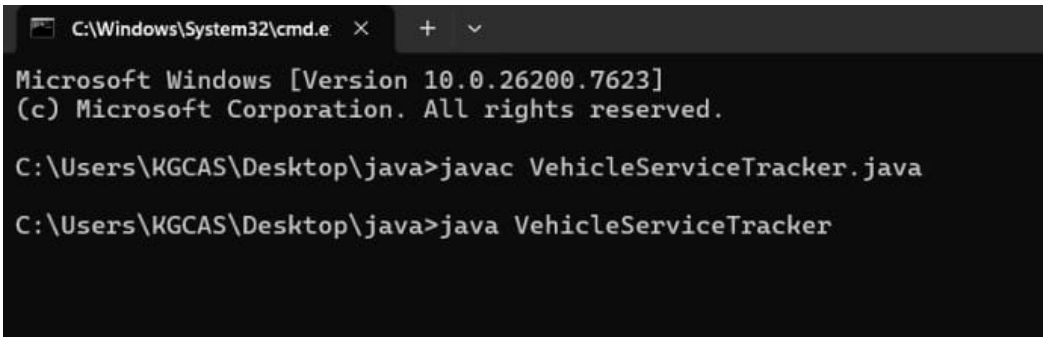
    public void serviceReminder() {
        System.out.println("\nReminder: Your next vehicle service is on " + nextServiceDate);
    }

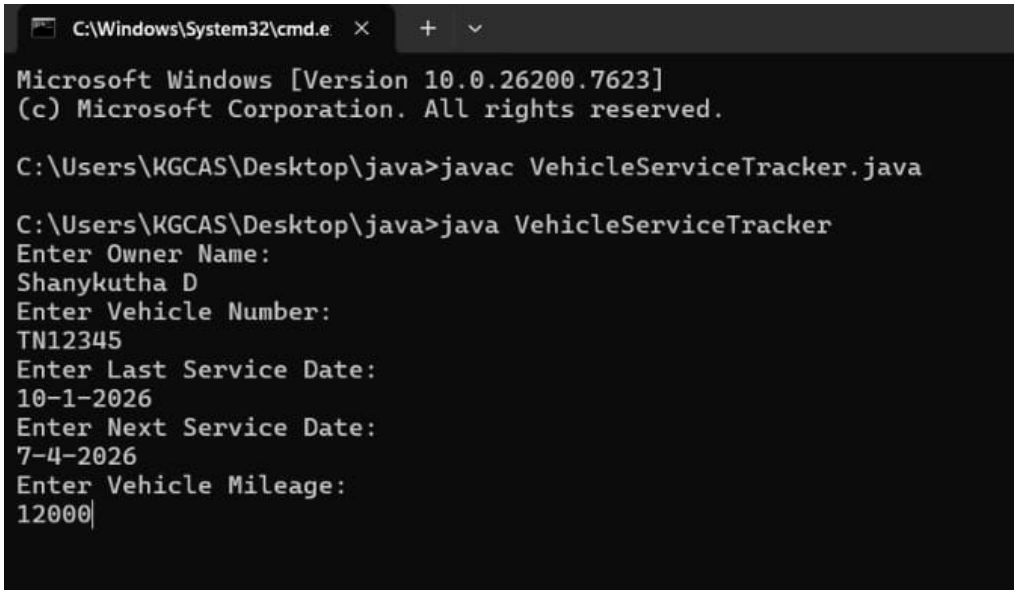
    public static void main(String[] args) {

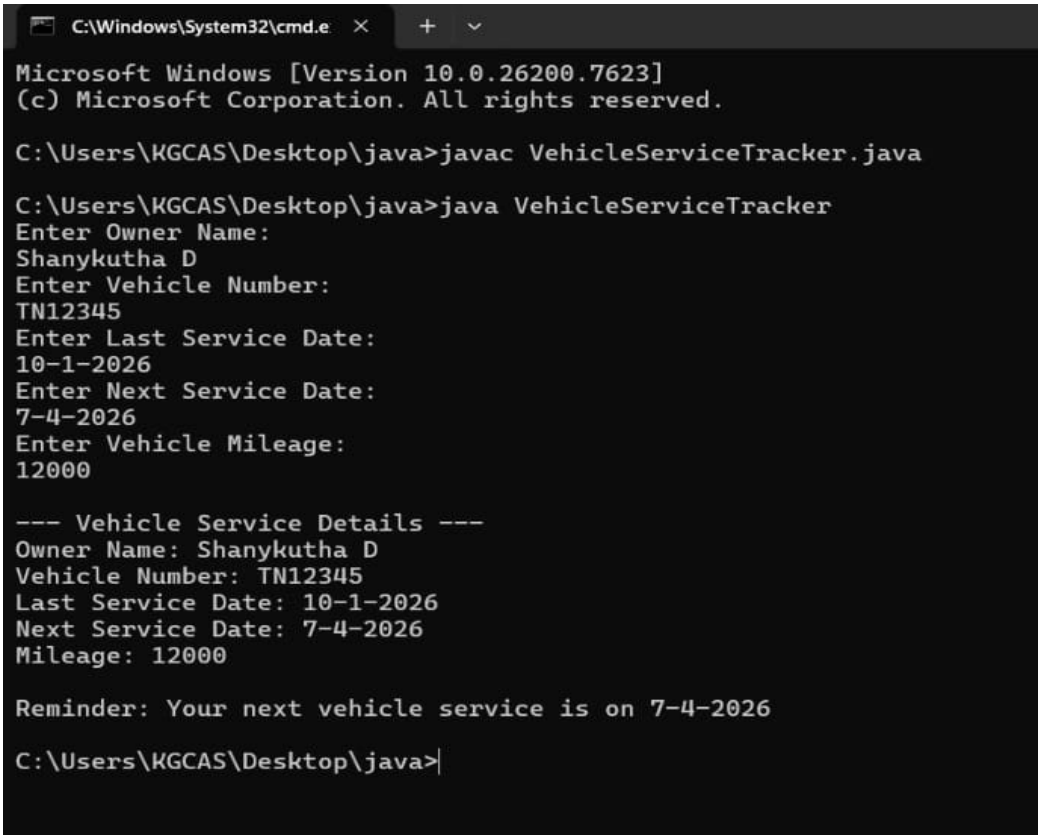
        VehicleServiceTracker v = new VehicleServiceTracker();

        v.getDetails();
        v.displayDetails();
        v.serviceReminder();
    }
}
```

6. INPUT AND OUTPUT SCREEN






C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.
C:\Users\KGCAS\Desktop\java>javac VehicleServiceTracker.java
C:\Users\KGCAS\Desktop\java>java VehicleServiceTracker
Enter Owner Name:
Shanykutha D
Enter Vehicle Number:
TN12345
Enter Last Service Date:
10-1-2026
Enter Next Service Date:
7-4-2026
Enter Vehicle Mileage:
12000
--- Vehicle Service Details ---
Owner Name: Shanykutha D
Vehicle Number: TN12345
Last Service Date: 10-1-2026
Next Service Date: 7-4-2026
Mileage: 12000
Reminder: Your next vehicle service is on 7-4-2026
C:\Users\KGCAS\Desktop\java>

7.DISCUSSION & CONCLUSION

7.1 Discussion

The Vehicle Service Reminder and Maintenance Tracker is designed to help users manage their vehicle service details efficiently. It stores important information such as vehicle number, service dates, and mileage. Many users forget their service schedules, which can affect vehicle performance. This system helps to overcome that issue by providing timely reminders.

The project is simple and user-friendly, making it suitable for all types of users. It improves awareness about regular vehicle maintenance. The system also helps in maintaining proper records for future reference. Users can easily access and update their vehicle details.

During development, it was observed that digital tracking is more reliable than manual methods. However, the system has some limitations, such as lack of automatic notifications and real-time data updates. Despite this, the project is effective for basic usage and learning purposes.

7.2 Conclusion

The Vehicle Service Reminder and Maintenance Tracker is a useful tool for maintaining vehicle service records. It helps users remember important service dates and ensures proper maintenance. The system is easy to use and reduces the chances of missing scheduled services.

This project demonstrates how technology can be used to solve everyday problems. It is reliable and efficient for basic vehicle tracking needs. Although it has some limitations, it provides a strong foundation for further improvements.

In the future, features like mobile application support, automatic alerts, and cloud storage can be added. Overall, this project is a practical and helpful solution for better vehicle maintenance and management.

8. REFERENCES

The following resources were used during the development of the Vehicle Service Reminder and Maintenance Tracker project.

Books

1.Java: The Complete Reference – Herbert Schildt

Websites:

1. Oracle Java Documentation – https://www.oracle.com/java
2. GeeksforGeeks – https://www.geeksforgeeks.org
3. W3Schools Java Tutorial – https://www.w3schools.com
4. Javatpoint Java Tutorial – https://www.javatpoint.com
5. TutorialsPoint Java – https://www.tutorialspoint.com